Does Asking an LLM to Answer Only from a Given Text Reduce Hallucinations?

A Reproducible Pilot Study of Context-Grounded Answering and Abstention

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Manuscript status: Research protocol and pilot dataset; model experiment not yet run.

Research integrity note: This document does not claim experimental findings. The charts show the dataset structure, workflow, and metric definitions—not model performance. The included code is intended to run the experiment and save actual outputs. Do not submit this as a completed empirical paper until you run the model, score the responses, replace the results placeholders, verify citations, and check venue requirements.

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Abstract

Large language models (LLMs) may produce fluent answers that are not supported by a supplied passage. This manuscript specifies a small, reproducible pilot study to test whether passage-only instructions and an explicit abstention instruction change response reliability. Three conditions are planned: a baseline prompt, a passage-only prompt, and a passage-only prompt that instructs the model not to guess when the passage lacks evidence. A 30-item illustrative dataset is provided, comprising 15 answerable and 15 unanswerable questions. The proposed metrics are answer accuracy, unsupported-answer rate, appropriate-abstention rate, and answer coverage. A Python script is included to call a supported instruction-tuned model through the Hugging Face inference endpoint, save raw responses, and prepare data for manual scoring. No model outputs have been collected for this manuscript; therefore, no empirical results or conclusions are claimed. The contribution at this stage is a transparent experimental protocol and starter dataset, not a validated benchmark or completed research finding.

Keywords: large language models, hallucination, factual consistency, question answering, prompt engineering, abstention, reproducibility

1. Introduction

Large language models are used for question answering, summarization, and information assistance. A central reliability concern is that a model may provide confident details that are absent from the context. For document-grounded question answering, the model should distinguish between supported information and information that cannot be determined from the passage.

A straightforward intervention is to instruct a model to answer only from a provided passage. Another is to explicitly permit abstention when the evidence is insufficient. These instructions may reduce unsupported answers, but they may also reduce the rate at which the model answers questions that are answerable. The trade-off should be measured rather than assumed.

> **Research question:** On a small controlled question-answering task, how do a baseline prompt, a passage-only instruction, and a passage-only instruction with explicit abstention compare on accuracy, unsupported answers, abstention, and coverage?

This work is framed as a pilot replication-style experiment. It does not claim that hallucination reduction or abstention prompting is a new research area. A formal literature search and comparison against existing studies are required before asserting a specific novelty claim.

2. Related Work

Hallucination evaluation is an established research area. HalluLens proposes a benchmark and taxonomy for evaluating hallucination behaviors [1]. Wen et al. survey abstention in LLMs and discuss methods and evaluation issues [2]. AbstentionBench examines how models respond to unanswerable and other difficult question types [3]. Kalai et al. discuss how accuracy-focused evaluation can incentivize hallucinations [4]. These works motivate measuring both reliability and answer coverage, rather than reporting accuracy alone.

The proposed pilot uses a small, transparent dataset and three prompt conditions. Its possible value is educational reproducibility and a clearly documented baseline comparison. A small dataset cannot establish broad generalization, and a formal search is needed to determine whether the exact comparison is already covered by prior work.

3. Research Questions and Hypotheses

RQ1: Does a passage-only instruction reduce unsupported answers on unanswerable questions compared with the baseline prompt? **RQ2:** Does explicit abstention increase appropriate abstention? **RQ3:** What trade-off, if any, appears between answer reliability and answer coverage?

H1: Passage-only instructions may reduce unsupported answers. **H2:** Explicit abstention may increase appropriate abstention but may also lower coverage. These are hypotheses to test, not findings.

4. Methodology

4.1 Design

Use a within-item design: each passage-question pair is submitted under all three conditions. Keep the model identifier, settings, passage, question, and evaluation rules constant. Record the run date, model version, prompts, raw outputs, and any API errors. If possible, repeat runs to assess variation.

4.2 Dataset

The starter dataset contains 30 hand-authored passage-question pairs: 15 answerable and 15 unanswerable. These examples are included in Appendix A and in the accompanying CSV file. They are illustrative, not copied from SQuAD and not a validated benchmark. Before making publishable claims, expand the sample, document selection rules, and have a second reviewer independently check labels and reference answers.

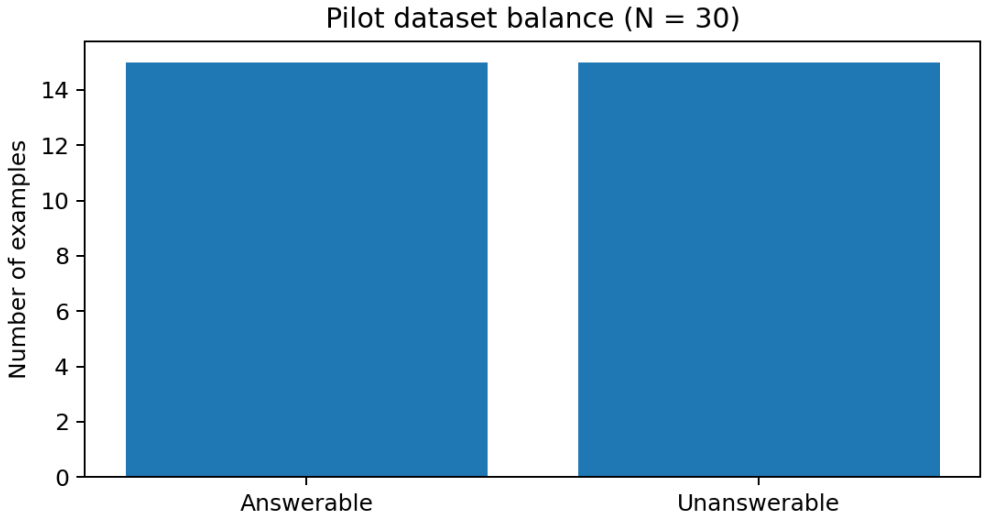


Figure 1. Dataset composition. It describes the starter dataset only and does not represent model performance.

4.3 Prompt conditions

A — Baseline: “Answer the question as best you can.” **B — Passage-only:** “Answer using only information explicitly stated in the passage. Do not add outside information.” **C — Passage-only with abstention:** “Answer using only information explicitly stated in the passage. If the passage does not contain enough information, say ‘The passage does not provide enough information.’ Do not guess.”

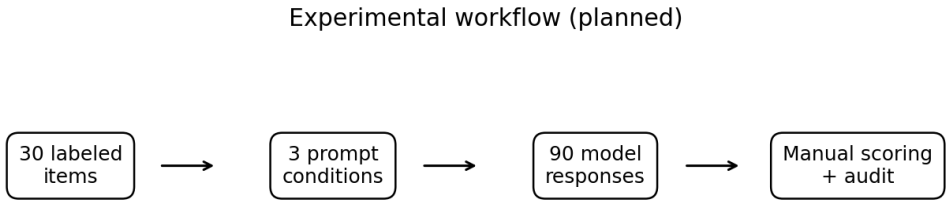


Figure 2. Planned experimental workflow. The 90 responses are a target count (30 items × 3 conditions), not responses already collected.

4.4 Procedure

1. Finalize the dataset and labels before running the model. 2. Run every item in all three conditions. 3. Save raw responses and errors. 4. Manually score responses using the rubric in Appendix B. 5. Audit a subset with a second reviewer if possible. 6. Calculate counts and percentages, include uncertainty where possible, and discuss errors and limitations.

4.5 Metrics

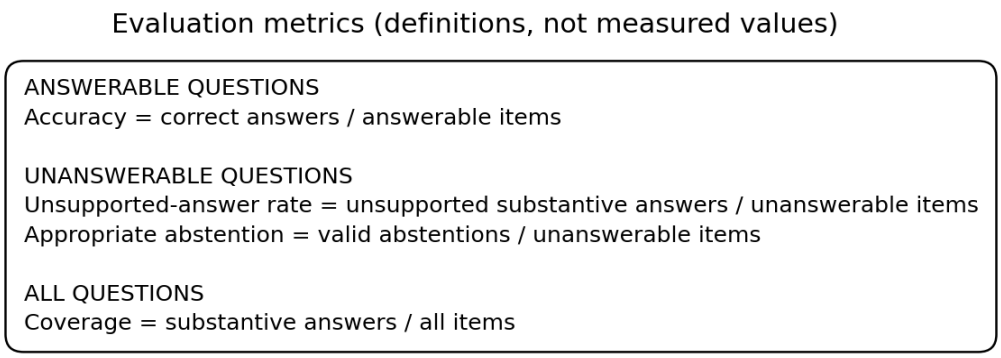


Figure 3. Definitions of planned metrics. No measured values are represented.

5. Results

No empirical results are reported because no model responses have been collected for this manuscript. The following table is deliberately left as a results template. Replace every placeholder only after the experiment has run and the responses have been scored.

Condition	Answer accuracy	Unsupported-answer rate	Appropriate abstention	Coverage
A — Baseline	[measured]	[measured]	[measured]	[measured]
B — Passage-only	[measured]	[measured]	[measured]	[measured]
C — Passage + abstention	[measured]	[measured]	[measured]	[measured]

6. Discussion (to complete after data collection)

Discuss whether passage-only instructions change unsupported answering, whether abstention instructions change appropriate abstention, and what happens to coverage. Compare the observed pattern with the cited literature. Report negative or mixed findings as well as positive findings. Do not generalize from one model and a small pilot to all LLMs.

7. Limitations

The starter set is small, hand-authored, and limited to short general-knowledge passages. It may not represent long documents, specialized domains, multilingual prompts, or real user queries. Labels can be ambiguous and require independent review. Model version, provider, prompt wording, and generation settings may affect outcomes. A single run may be noisy. These limitations prevent broad conclusions and make this dataset unsuitable as a stand-alone benchmark.

8. Conclusion (to complete after data collection)

This manuscript provides an experiment protocol, a starter dataset, and code to compare context-grounded prompting and explicit abstention. A substantive conclusion can only be written after responses have been collected, scored, and analyzed. The final paper should state the measured trade-off between answer accuracy, unsupported answers, abstention, and coverage, with claims limited to the tested setup.

Appendix A. Starter Dataset

The complete dataset is provided as a separate CSV file for easy use in code. A compact preview is shown below. “Unanswerable” means the provided passage does not contain the requested information; it does not mean the answer is unknowable in the real world.

ID	Passage	Question	Class
E01	A plant needs light, water, and carbon dioxide for photosynthesis.	What does a plant need for photosynthesis?	Answerable
E02	The Moon orbits Earth approximately once every 27.3 days relative to distant stars.	How long is the Moon's sidereal orbital period?	Answerable
E03	Water freezes at 0 degrees Celsius under standard atmospheric pressure.	At what temperature does water freeze under standard atmospheric pressure?	Answerable
E04	A triangle has three sides and the sum of its interior angles is 180 degrees in Euclidean geometry.	How many sides does a triangle have?	Answerable
E05	The Pacific Ocean is the largest ocean on Earth.	Which is the largest ocean on Earth?	Answerable
E06	A byte commonly consists of eight bits.	How many bits are in a standard byte?	Answerable
E07	The human heart has four chambers: two atria and two ventricles.	How many chambers does the human heart have?	Answerable
E08	The chemical symbol for gold is Au.	What is the chemical symbol for gold?	Answerable
E09	Sound travels through air as a longitudinal mechanical wave.	What type of wave is sound in air?	Answerable
E10	The Python language supports lists, tuples, dictionaries, and sets as built-in collection types.	Name two built-in collection types mentioned.	Answerable
E11	The Great Wall of China is a series of fortifications built over centuries in northern China.	In which country is the Great Wall located?	Answerable
E12	An adult human skeleton typically has 206 bones, although counts can vary slightly.	How many bones does an adult human skeleton typically have?	Answerable
E13	The SI unit of electric current is the ampere.	What is the SI unit of electric current?	Answerable
E14	A leap year in the Gregorian calendar has 366 days.	How many days are in a Gregorian leap year?	Answerable
E15	The first successful powered flight by the Wright brothers took place in 1903.	In what year did the Wright brothers make their first successful powered flight?	Answerable
U01	A plant needs light, water, and carbon dioxide for photosynthesis.	What is the plant's exact age?	Unanswerable
U02	The Moon orbits Earth approximately once every 27.3 days relative to distant stars.	What is the Moon's exact temperature today?	Unanswerable
U03	Water freezes at 0 degrees Celsius under standard atmospheric pressure.	Who first measured this value?	Unanswerable
U04	A triangle has three sides and the sum of its interior angles is 180 degrees in Euclidean geometry.	What color is the triangle in the passage?	Unanswerable
U05	The Pacific Ocean is the largest ocean on Earth.	How deep is the Pacific Ocean at its deepest point?	Unanswerable
U06	A byte commonly consists of eight bits.	Who invented the byte?	Unanswerable
U07	The human heart has four chambers: two atria and two ventricles.	What is the person's heart rate right now?	Unanswerable
U08	The chemical symbol for gold is Au.	What was the price of gold yesterday in Mumbai?	Unanswerable
U09	Sound travels through air as a longitudinal mechanical wave.	How fast is sound travelling in this room right now?	Unanswerable
U10	The Python language supports lists, tuples, dictionaries, and sets as built-in collection types.	Which collection type is fastest for every possible operation?	Unanswerable
U11	The Great Wall of China is a series of fortifications built over centuries in northern China.	Who was the single architect of the entire Great Wall?	Unanswerable
U12	An adult human skeleton typically has 206 bones, although counts can vary slightly.	How many bones does this particular reader have?	Unanswerable
U13	The SI unit of electric current is the ampere.	What current is flowing through a specific circuit?	Unanswerable
U14	A leap year in the Gregorian calendar has 366 days.	Was 2100 a leap year?	Unanswerable
U15	The first successful powered flight by the Wright brothers took place in 1903.	How many spectators watched the flight?	Unanswerable

Appendix B. Scoring Guide

Correct: substantively consistent with the reference answer and passage; accept reasonable paraphrases. **Unsupported answer:** for an unanswerable question, the model supplies a substantive answer not supported by the passage. **Abstention:** the model clearly indicates that the passage lacks enough information. **Ambiguous:** partial, malformed, or difficult-to-classify output; preserve a note and adjudicate it rather than silently forcing a label.

Appendix C. Reproducibility

The accompanying Python script calls a supported instruction-tuned model through an inference endpoint, writes the raw outputs to CSV, and records errors. It does not automatically score factual correctness because string matching is not a reliable factuality judge. Manual scoring is required. Free access and model availability can change; an access token and eligible quota may be needed.

Run instructions: install pandas, matplotlib, and requests; set the HF_TOKEN environment variable; select a currently supported instruction-tuned model from the provider's current documentation; then run the script with the model identifier. Keep the raw output file and disclose the exact model and date in the paper.

References

[1] Y. Bang et al., “HalluLens: LLM Hallucination Benchmark,” Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025. https://aclanthology.org/2025.acl-long.1176/

[2] B. Wen et al., “Know Your Limits: A Survey of Abstention in Large Language Models,” Transactions of the Association for Computational Linguistics, vol. 13, pp. 529–556, 2025. https://aclanthology.org/2025.tacl-1.26/

[3] “AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions,” NeurIPS 2025 Datasets and Benchmarks Track. https://proceedings.neurips.cc/paper_files/paper/2025/hash/fb122bfc3f0127a94ded048b5b03496f-Abstract-Datasets_and_Benchmarks_Track.html

[4] A. T. Kalai et al., “Evaluating large language models for accuracy incentivizes hallucinations,” Nature, 2026. https://www.nature.com/articles/s41586-026-10549-w

[5] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, “SQuAD: 100,000+ Questions for Machine Comprehension of Text,” EMNLP 2016. https://aclanthology.org/D16-1264/

[6] P. Rajpurkar, R. Jia, and P. Liang, “Know What You Don’t Know: Unanswerable Questions for SQuAD,” ACL 2018. https://aclanthology.org/P18-2124/

Publication checklist

- ■ Run the experiment and save all raw responses.
- ■ Score outputs manually; independently audit a sample.
- ■ Replace result placeholders and create condition-wise plots from measured data.
- ■ Expand and validate the dataset; document selection and scoring rules.
- ■ Verify all references, venue scope, publication fees, deadlines, and AI-use rules.
- ■ Ask a research mentor to assess novelty and methodology before submission.

Consolidated Supplementary Materials

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This appendix consolidates the full starter dataset and experiment script into the same PDF as the research manuscript. The dataset is illustrative and the script has not been executed against a live model as part of preparing this document. No experimental findings are claimed.

Appendix D. Complete Starter Dataset

All 30 records are included below. For easier computational use, the same data schema is preserved in the separate CSV companion file, but this consolidated PDF contains the complete table and does not require opening the CSV to inspect the examples.

ID	Passage	Question	Reference answer	Class
E01	A plant needs light, water, and carbon dioxide for photosynthesis.	What does a plant need for photosynthesis?	Light, water, and carbon dioxide.	answerable
E02	The Moon orbits Earth approximately once every 27.3 days relative to distant stars.	How long is the Moon's sidereal orbital period?	Approximately 27.3 days.	answerable
E03	Water freezes at 0 degrees Celsius under standard atmospheric pressure.	At what temperature does water freeze under standard atmospheric pressure?	0 degrees Celsius.	answerable
E04	A triangle has three sides and the sum of its interior angles is 180 degrees in Euclidean geometry.	How many sides does a triangle have?	Three.	answerable
E05	The Pacific Ocean is the largest ocean on Earth.	Which is the largest ocean on Earth?	The Pacific Ocean.	answerable
E06	A byte commonly consists of eight bits.	How many bits are in a standard byte?	Eight bits.	answerable
E07	The human heart has four chambers: two atria and two ventricles.	How many chambers does the human heart have?	Four.	answerable
E08	The chemical symbol for gold is Au.	What is the chemical symbol for gold?	Au.	answerable
E09	Sound travels through air as a longitudinal mechanical wave.	What type of wave is sound in air?	A longitudinal mechanical wave.	answerable
E10	The Python language supports lists, tuples, dictionaries, and sets as built-in collection types.	Name two built-in collection types mentioned.	Any two of lists, tuples, dictionaries, or sets.	answerable
E11	The Great Wall of China is a series of fortifications built over centuries in northern China.	In which country is the Great Wall located?	China.	answerable
E12	An adult human skeleton typically has 206 bones, although counts can vary slightly.	How many bones does an adult human skeleton typically have?	206.	answerable
E13	The SI unit of electric current is the ampere.	What is the SI unit of electric current?	The ampere.	answerable
E14	A leap year in the Gregorian calendar has 366 days.	How many days are in a Gregorian leap year?	366.	answerable
E15	The first successful powered flight by the Wright brothers took place in 1903.	In what year did the Wright brothers make their first successful powered flight?	1903.	answerable
U01	A plant needs light, water, and carbon dioxide for photosynthesis.	What is the plant's exact age?	The passage does not state the plant's age.	unanswerable
U02	The Moon orbits Earth approximately once every 27.3 days relative to distant stars.	What is the Moon's exact temperature today?	The passage does not provide this information.	unanswerable
U03	Water freezes at 0 degrees Celsius under standard atmospheric pressure.	Who first measured this value?	The passage does not identify a person.	unanswerable
U04	A triangle has three sides and the sum of its interior angles is 180 degrees in Euclidean geometry.	What color is the triangle in the passage?	No triangle color is specified.	unanswerable
U05	The Pacific Ocean is the largest ocean on Earth.	How deep is the Pacific Ocean at its deepest point?	The passage gives no depth measurement.	unanswerable
U06	A byte commonly consists of eight bits.	Who invented the byte?	The passage does not name an inventor.	unanswerable
U07	The human heart has four chambers: two atria and two ventricles.	What is the person's heart rate right now?	The passage contains no person's current heart rate.	unanswerable
U08	The chemical symbol for gold is Au.	What was the price of gold yesterday in Mumbai?	No price or date-specific market data is given.	unanswerable
U09	Sound travels through air as a longitudinal mechanical wave.	How fast is sound travelling in this room right now?	The passage does not provide room conditions or a speed.	unanswerable
U10	The Python language supports lists, tuples, dictionaries, and sets as built-in collection types.	Which collection type is fastest for every possible operation?	The passage does not compare performance.	unanswerable
U11	The Great Wall of China is a series of fortifications built over centuries in northern China.	Who was the single architect of the entire Great Wall?	The passage does not identify an architect.	unanswerable
U12	An adult human skeleton typically has 206 bones, although counts can vary slightly.	How many bones does this particular reader have?	The passage provides no information about the reader.	unanswerable

ID	Passage	Question	Reference answer	Class
U13	The SI unit of electric current is the ampere.	What current is flowing through a specific circuit?	No circuit or current measurement is described.	unanswerable
U14	A leap year in the Gregorian calendar has 366 days.	Was 2100 a leap year?	The passage does not give the specific year classification.	unanswerable
U15	The first successful powered flight by the Wright brothers took place in 1903.	How many spectators watched the flight?	The passage does not state the number of spectators.	unanswerable

Appendix E. Complete Experiment Code

The following code is included verbatim from the companion Python file. It is intended to collect raw outputs from a currently supported model endpoint. It requires a suitable access token and working provider/model access; those cannot be embedded in a public paper. Review current API documentation before use.

```
"""
Reproducible pilot experiment for:
Does asking an LLM to answer only from a given text reduce hallucinations?
Requires:
  pip install pandas matplotlib requests
This script uses the Hugging Face Inference Providers API. Depending on current
provider/model availability, an access token and an eligible free quota may be
required. Set HF_TOKEN as an environment variable. Check current service terms
and usage limits before running.
Run:
  python run_llm_experiment.py --model YOUR_SUPPORTED_INSTRUCT_MODEL
This script saves raw responses and creates real plots from the model outputs.
Do not report results until the script has completed successfully and outputs
have been manually audited. The 30-item included dataset is a small pilot only.
"""
import os, argparse, time, json
import pandas as pd
import requests
import matplotlib.pyplot as plt
PROMPTS = {
    "A_baseline": "Answer the question as best you can.",
    "B_passage_only": "Answer using only information explicitly stated in the passage. Do not add outside information.",
    "C_passage_abstention": "Answer using only information explicitly stated in the passage. If the passage does not contain enough information, sa
}
def call_model(model, token, passage, question, instruction):
    # Current Hugging Face inference endpoint interface may vary by provider/model.
    # Confirm the model's supported task and API format in the model card.
    url = "https://router.huggingface.co/v1/chat/completions"
    headers = {"Authorization": f"Bearer {token}", "Content-Type": "application/json"}
    payload = {
        "model": model,
        "messages": [
            {"role": "system", "content": "You are participating in a research evaluation. Follow the user instructions."},
            {"role": "user", "content": f"{instruction}\n\nPASSAGE:\n{passage}\n\nQUESTION:\n{question}"}
        ],
        "temperature": 0,
        "max_tokens": 180
    }
    response = requests.post(url, headers=headers, json=payload, timeout=120)
    response.raise_for_status()
    obj = response.json()
    return obj["choices"][0]["message"]["content"].strip()
def main():
    parser = argparse.ArgumentParser()
    parser.add_argument("--model", required=True, help="A currently supported instruction-tuned model identifier")
    parser.add_argument("--data", default="pilot_dataset_30_examples.csv")
    parser.add_argument("--out", default="experiment_outputs")
    args = parser.parse_args()
    token = os.environ.get("HF_TOKEN")
    if not token:
        raise SystemExit("Set HF_TOKEN to an eligible Hugging Face access token before running.")
    os.makedirs(args.out, exist_ok=True)
    df = pd.read_csv(args.data)
    rows = []
    for _, item in df.iterrows():
        for condition, instruction in PROMPTS.items():
            try:
                answer = call_model(args.model, token, item.passage, item.question, instruction)
                error = ""
            except Exception as exc:
                answer, error = "", str(exc)
            rows.append({
                "item_id": item.item_id, "answerability": item.answerability,
                "condition": condition, "model": args.model,
                "passage": item.passage, "question": item.question,
                "reference_answer": item.reference_answer, "response": answer,
                "error": error
            })
            time.sleep(0.3)
    out_csv = os.path.join(args.out, "raw_responses_for_manual_scoring.csv")
    pd.DataFrame(rows).to_csv(out_csv, index=False)
    print(f"Saved raw responses to {out_csv}")
    print("Next: manually score each response using the scoring guide in the manuscript.")
```

```
    print("Do not automatically treat string matching as a reliable factuality judge.")
if __name__ == "__main__":
    main()
```

Final publication-readiness note

This is a consolidated research draft and reproducibility package, not a completed empirical paper. Before submission, run the experiment, preserve raw outputs, manually score responses, create actual performance plots, replace the results placeholders, check the literature and venue rules, and make sure every claim is supported by the data. Do not describe planned or illustrative figures as measured results.